

Genetic gap indicators: the short proof

Research note • 20 September 2026

Targets. Conjecture 53 of Rubinstein-Salzedo–Swaminathan [1] asserts uniqueness, modulo constants, of genetic primitives. Chapter 8, Conjecture 3 of Berenbeim [2] asserts the sup property for entire genetic functions. Here “genetic” uses their set-sized option constructors and the conventional treatment of option expressions that ignore the options of their input.

Construction. With $\mathbb{N} = \{0, 1, 2, \dots\}$ the ordinary finite naturals, define

$$q(t) = \frac{t}{1+t^2}, \quad T(x) = \left\{ q(n-x) : n \in \mathbb{N} \mid \right\}, \quad B(x) = 1 - T(x).$$

The denominator is positive, $q(t)$ has the sign of t , and $|q(t)| \leq 1/2$, since $(|t| - 1)^2 \geq 0$.

Claim. $B(x) = 1$ exactly when $x > n$ for every $n \in \mathbb{N}$, and is 0 otherwise.

Proof. If $x > n$ for all n , every left option of $T(x)$ is negative. Thus 0, the simplest surreal, lies above all of them, giving $T(x) = 0$. Otherwise choose $n \geq x$. Then one left option is nonnegative, excluding 0, while every left option is less than 1. The simplest admissible number is now 1, so $T(x) = 1$. Subtract from 1. $\square$

Geneticity, including the later cofinality condition. The globally defined rational q is genetic; the sources explicitly use the one-option cut $\{ q(x) \mid \}$. Each $q(n-x)$ is an admissible affine translate of a previously constructed function. The option set is fixed and countable, and the right option set is empty. Thus T is a nonrecursive genetic adjunction. No input is excluded.

The options reference neither x^L nor x^R , so changing the representation of x leaves the cut unchanged. For every $y < x < z$, the later global cofinality requirement still reads $q(n-x) < T(x)$. If $T(x) = 1$ this follows from $q(n-x) \leq 1/2$; if $T(x) = 0$ every option is negative. The dual condition is vacuous. Arithmetic closure makes B and $x - B(x)$ genetic as well.

Derivative. For every surreal x and every h with $|h| < 1$, membership in the positive infinite class is unchanged by adding h . Indeed, an integer upper bound remains an integer upper bound after increasing it by 1; and $x > n + 1$ for all n implies $x + h > n$ for all n . Consequently

$$B(x+h) = B(x), \quad \frac{B(x+h) - B(x)}{h} = 0 \quad (0 < |h| < 1).$$

Thus $B' = 0$ everywhere in the full surreal ε - δ sense, although $B(0) = 0$ and $B(\omega) = 1$.

First counterexample. Set $F_0(x) = x$ and $F_1(x) = x - B(x)$. For $0 < |h| < 1$, the quotient for F_1 is exactly 1, so

$$F'_0 = F'_1 = 1, \quad F_1 - F_0 = -B \text{ is nonconstant.}$$

Both functions agree with x at every real argument. The same counterexample works on the proper bounded open interval $(0, \omega)$, using the points 1 and $\omega/2$ to witness nonconstant difference. At each interior point shrink the derivative neighborhood to stay in the interval.

Moreover, F_1 is an increasing bijection. It is the identity below the finite–infinite gap and translation by -1 above it. Translation by a finite number preserves both sides. Any cross-gap difference is positive infinite, so subtracting 1 preserves its sign. The inverse is $x + B(x)$. The piecewise form of F_1 already appears in [1, p. 37, eq. (7)]; the additional content here is its genetic formula.

Second counterexample. Consider the sublevel class

$$E = \{x \in \mathbf{No} : 0 < x < \omega, B(x) \leq 1/2\}.$$

It is exactly the positive finite surreals. Its surreal upper bounds are exactly the positive infinite surreals: it contains every positive integer, and each of its members is bounded by an integer. Every such upper bound u can be replaced by the smaller upper bound $u/2$. Thus E has no surreal least upper bound. It is nonempty and bounded by ω , so neither end gap is its supremum. The sup property fails.

Why there is no missing derivative point. The gap between numbers bounded above by a natural and numbers above all naturals is not itself a surreal number. Its lower side has no greatest element and its upper side has no least element. A derivative is computed at an actual surreal input, never at this unrealized boundary. In particular, ω is on the upper side, not at the boundary.

Scope and validation. These are counterexamples under the published set-sized option conventions, not a claim about every more restrictive use of the term “genetic.” No later resolution was located in the recorded searches, but priority is not established. Independent expert review and proof-assistant verification have not been performed. The full article contains the general set-indexed construction, ordinal-indexed independent families, a definition audit, and exact finite computational checks with their limitations.

References

- [1] S. Rubinstein-Salzedo and A. Swaminathan, *Analysis on Surreal Numbers*, Journal of Logic & Analysis 6:5 (2014), 1–39. Constructor and examples: pp. 9–11; Conjecture 53: p. 37. <https://doi.org/10.4115/jla.2014.6.5>.
- [2] A. M. Berenbeim, *Foundations for the Analysis of Surreal-Valued Genetic Functions*, Ph.D. dissertation, University of Illinois at Chicago, 2022. Definition 55: pp. 126–128; rational step example: pp. 133–134; Chapter 8, Conjecture 3: pp. 225–226. [UIC repository PDF \(version 1\)](#).